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Assignment No. - 1

Applied Physics

Derivation of Maxwell's Equations

(1) In Differential Form :

(i) " Gauss Law of electricity " : $\nabla . \vec{D} = \rho$

Let us consider a surface S bounding total charge consisting of free charge plus polarisation. If ρ and ρ_P are the charge densities of free charge and polarization charge at a point in a small volume element dV, then Gauss's Law can be expressed as :-

$$\oint \vec{E}.\vec{ds} = \frac{q}{\epsilon_0} = \phi_E \quad -(1)$$

$$\oint \vec{E}.\vec{ds} = \frac{1}{\epsilon_0}\int \rho . dV \quad -(2)$$

$$\oint \vec{E}.\vec{ds} = \frac{1}{\epsilon_0}\int (\rho + \rho_P)\, dV$$

But we know $\rho_P = -\operatorname{div} \vec{P}$ where $\vec{P} \rightarrow$ Polarization Vector

$$\oint \vec{E}.\vec{ds} = \frac{1}{\epsilon_0}\int (\rho - \operatorname{div}. \vec{P}) . dV$$

$$\epsilon_0 . \oint_S \vec{E}.\vec{ds} = \int \rho . dV - \int_V (\operatorname{div} \vec{P}) . dV$$

Applying Gauss Divergence Theorem :

$$\oint \operatorname{div}(\epsilon_0 \vec{E}) . dV = \int \rho . dV - \int (\operatorname{div} \vec{P}) . dV$$

$$\oint \operatorname{div}. (\epsilon_0 \vec{E} + \vec{P})\, dV = \int \rho . dV$$

$$\oint (\operatorname{div}. \vec{D} . dV) = \int \rho . dV$$

$$\oint_V [\operatorname{div} \vec{D} - \rho]\, dV = 0$$

For arbitrary volume, this integral vanishes.

$$\operatorname{div}. \vec{D} - \rho = 0$$

$$\boxed{\nabla . \vec{D} = \rho} \rightarrow \text{Maxwell's first equation of electromagnetism.}$$

(ii) " Based on Gauss Law of Magnetism" : $\nabla . \vec{B} = 0$

The number of magnetic lines of forces entering any arbitrary closed surface is exactly the same as leaving it. It means that the flux of magnetic induction B across the closed surface is always zero.

$$\int_S \vec{B}.\vec{ds} = 0$$

using Gauss's Divergence Theorem to change surface integral into volume integral:

$$\int_V (\text{div } \vec{B})\, dV = 0$$

For arbitrary volume, this integral vanishes:

$$\text{div } \vec{B} = 0$$

$$\boxed{\nabla . \vec{B} = 0}$$ → Maxwell's second equation of electromagnetism

Conclusion:- Thus, isolated magnetic poles and magnetic currents due to them have no physical significance : therefore magnetic lines of force in general are either closed curves or go off to infinity.

(iii) " Based on Faraday's Law of electromagnetic induction":

$$\nabla \times \vec{E} = -\frac{\delta \vec{B}}{\delta t}$$

Acc. to Faraday's Law of electromagnetic Induction, emf induced in a closed loop is, $e = -\frac{d\phi_B}{dt}$ —①

Magnetic Flux ⇒ $\phi_B = \oint_S \vec{B}.\vec{ds}$ ⇒ putting in eq.①

$$\Rightarrow \quad e = -\frac{d}{dt}\left(\oint \vec{B}.\vec{ds}\right)$$

$$e = \oint_l \vec{E}.\vec{dl} = -\frac{d}{dt}\left(\oint \vec{B}.\vec{ds}\right)$$

$$e = -\oint_S \frac{\delta \vec{B}}{\delta t} . \vec{ds}$$ (∵ surface is fixed in space, hence only B changes with time)

Applying Stoke's Theorem:

$$\oint (\nabla \times \vec{E}).\vec{ds} = -\oint \frac{\delta \vec{B}}{\delta t} . \vec{ds}$$

$$\oint \left(\nabla \times \vec{E} + \frac{\delta \vec{B}}{\delta t}\right) \cdot \vec{ds} = 0$$

For arbitrary surface, this integral vanishes.

$$\boxed{\nabla \times \vec{E}' = -\frac{\delta \vec{B}}{\delta t}} \longrightarrow$$ Maxwell's third equation of electromagnetism

(iv) "Based on Ampere's Law for currents in conductors":

$$\nabla \times H = \vec{J} + \frac{\delta \vec{D}}{\delta t}.$$

Ampere's circuital Law: $\oint H \cdot dl = I = \oint J \cdot \vec{ds}$

where I is total current i.e. sum of conduction + Displacement current

- Apply Stoke's theorem to convert line integral into surface Integral:

$$\oint (\nabla \times H) ds = \oint \vec{J} \cdot ds$$

Ampere's Law is valid for only steady currents only and not for varying fields. Thus, it is inconsistent and needs modification.

$$\oint [(\nabla \times H - \vec{J}] \cdot \vec{ds} = 0$$

$$\nabla \times H = \vec{J} \quad -(1)$$

$\therefore$ eq. (1) is inconsistent.

$$\nabla \times \vec{H} = \vec{J} + \vec{J}' \quad -(2)$$

$\therefore$ eq. (2) is valid only for time-varying fields.

- Taking Divergence on both the sides of eq. (2):

$$\text{div.}(\text{curl } \vec{H}) = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}'$$

($\because$ Div. (curl of vector quantity) is always zero]

$$\nabla \cdot \vec{J} + \nabla \cdot \vec{J}' = 0$$

$$\nabla \cdot \vec{J} = -\nabla \cdot \vec{J}'$$

using continuity equation: $\nabla \cdot \vec{J} = -\frac{\delta \rho}{\delta t}$

$$\frac{\delta \rho}{\delta t} = \nabla \cdot \vec{J}$$

$$\nabla \cdot D = \rho$$

$$\frac{\delta (\nabla \cdot D)}{\delta t} = \nabla \cdot \vec{J}$$

$$\Rightarrow \nabla \cdot \vec{J}' = \nabla \cdot \left(\frac{\delta D}{\delta t}\right)$$

$$\Rightarrow \vec{J}' = \frac{\delta D}{\delta t}$$

putting value of $\vec{J}''$ in eq. ②:

$$\boxed{\nabla \times \vec{H} = \vec{J}' + \frac{\delta D}{\delta t}}$$ → Maxwell's fourth Equation

(2) In Integral Form:

(i) Maxwell's First Equation: $\boxed{\nabla . D = \rho}$ – eq. ①

Integrating eq. ① over an arbitrary volume V:

$$\int_V \nabla . D . dV = \int \rho . dV$$

Applying Gauss Divergence Theorem:

$$\boxed{\int_S D . ds = \int_V \rho . dV}$$ → Maxwell's first equation in Integral form.

where S is the surface which bounds Volume V.

$\int \rho . dV = q$ (net charge contained in volume V)

(ii) Maxwell's Second Equation: $\boxed{\nabla . B = 0}$

Integrating this over an arbitrary volume V:

$$\int_V \nabla . B \, dV = 0$$

Using Gauss Divergence theorem to change volume integral into surface integral ⇒ $\boxed{\int_S B . ds = 0}$ → Maxwell's second equation in integral form.

where S is the surface, which bounds volume V.

(iii) Maxwell's Third Equation: $\boxed{\nabla \times E = -\frac{\delta B}{\delta t}}$

Integrating above equation over a surface S bounded by a curve C:

$$\int_S (\nabla \times E) . dS = -\int_S \frac{\delta B}{\delta t} . dS$$

Using Stoke's Theorem: $\boxed{\int_l E . dl = -\frac{\delta}{\delta t} \int_S B . dS}$ → Maxwell's third equation in integral form.

(iv) Maxwell's Fourth Equation : $\nabla \times H = J + \frac{\delta D}{\delta t}$

Taking surface integral over surface S bounded by curve C :

$$\int (\nabla \times H) ds = \int \left(J + \frac{\delta D}{\delta t}\right) . ds$$

using Stoke's theorem to convert surface integral to line integral ⇒

$$\boxed{\oint H . dl = \int_s \left(J + \frac{\delta D}{\delta t}\right) . ds} \rightarrow$$

Maxwell's fourth equation in integral form.